

Exercise Set 4 - Part 1

Problem 1

- 1.1 Give two advantages of panel data.
- 1.2 Give two limitations of panel data.
- 1.3 What are the two basic approaches to model unobserved effects in panel data? How do the two approaches differ (assumptions, advantages/disadvantages)?

Problem 2

Using the Youth Sample of the National Longitudinal Survey, log wages are explained from years of schooling, years of experience and its square, dummy variables for being a union member, working in the public sector and being married and two racial dummies. The estimation results under three different models are reported below.

	OLS	Fixed Effects	Random Effects
schooling	0.099 (0.005)	- -	0.101 (0.009)
experience	0.089 (0.010)	0.116 (0.008)	0.112 (0.008)
<i>experience</i> ²	-0.0028 (0.0007)	- 0.0043 (0.0006)	-0.0041 (0.0006)
union member	0.180 (0.017)	0.081 (0.019)	0.106 (0.018)
married	0.108 (0.016)	0.045 (0.018)	0.063 (0.017)
black	-0.144 (0.024)	- -	-0.144 (0.048)
hispanic	0.016 (0.021)	- -	0.020 (0.043)
public sector	0.004 (0.037)	0.035 (0.039)	0.030 (0.036)
constant	-0.034 (0.065)	- -	-0.104 (0.111)

Table 1: Dependent variable: log(wage), Standard errors are in parentheses

- 2.1 If the assumptions of the random effects model are satisfied, which estimators in Table 1 are consistent? Which estimator is the most efficient one?
- 2.2 If the individual-specific error term component is correlated with one or more of the explanatory variables, which estimators in Table 1 are consistent? Which test would you perform to check this hypothesis?
- 2.3 Show the within transformation of the panel data model $y_{it} = \alpha_i + \mathbf{x}'_{it}\boldsymbol{\beta} + \varepsilon_{it}$. What assumption is necessary to make the fixed effects estimator consistent?
- 2.4 Compare the results of fixed effects and random effects model. Interpret the coefficient of being married.

Problem 3

Utilizing panel data on individuals from the NLSY, we will analyze the impact of education on log wages. We include dummy-coded variables for region, race, sex and marital status as well as experience (number of years in the workforce) and tenure (number of years at the specific job), both also squared.

- 3.1 Set data as panel data.
- 3.2 Run a pooled OLS regression. What is the return of education? Assess whether it is statistically significant.
- 3.3 Include individual-level fixed effects.
- 3.4 Estimate a random effects model.
- 3.5 Interpret and compare the estimates of the three estimated models.
- 3.6 Run a Hausman test for fixed versus random effects model. What is your conclusion?